

**Department of Chemical & Biomolecular Engineering
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CHE330 Convection and Radiation Experiment*

Objective

To study, analyze, and compare the extents by which free convection, forced convection, and radiation contribute to the heat transfer from a heated cylinder under various temperature and air flow conditions.

Background and Theory

(1) Natural or Free Convection

When a hot object is suspended in stationary ambient fluid (which can be gas or liquid), heat will be lost from the hot object via natural or free convection. Natural convection occurs because hot fluid, being less dense generally, will rise above cooler fluid that is generally denser. Thus, the fluid immediately surrounding a hot object will get heated and will rise and be displaced by cooler ambient fluid. The resulting fluid circulation will carry heat away from the hot object and dissipate it to the rising fluid. The heat transfer rate from the hot object via natural convection, Q_{nc} , is given by:

$$Q_{nc} = H_{nc} A_s (T_s - T_a) \quad (1)$$

where H_{nc} is the natural convection heat transfer coefficient,

T_s is the surface temperature of the hot object,

T_a is the temperature of the ambient fluid away from the hot object,

A_s is the heat transfer surface area of the hot object.

For a hot metal cylinder of diameter D that is suspended in ambient air, which is the special case of interest in this experiment, H_{nc} is given by the following simplified empirical equation [1]:

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$$H_{nc}(W/m^2K) = 1.32[(T_s - T_a)/D]^{0.25} \quad (2)$$

(2) Forced Convection

When a mechanical means such as a fan, blower, or pump is used to drive a fluid over a hot object, heat is transferred from the hot object to the fluid flowing over it, and the mode of heat transfer is known as forced convection. The heat transfer rate from the hot object via forced convection, Q_{fc} , is analogous to Equation 1:

$$Q_{fc} = H_{fc} A_s (T_s - T_a) \quad (3)$$

where H_{fc} is the forced convection heat transfer coefficient and the other parameters are as defined previously. H_{fc} can be estimated from an empirical correlation for the Nusselt number defined for the forced convection system, i.e.

$$\frac{H_{fc} D}{K} \equiv Nu = 0.3 + \frac{0.62 Re^{0.5} Pr^{0.33}}{[1 + (\frac{0.4}{Pr})^{0.66}]^{0.25}} \left[1 + \left(\frac{Re}{282000} \right)^{0.5} \right] \quad (4)$$

This correlation was developed for forced convection of fluid over a straight cylinder of diameter D [2]. In the correlation, Re and Pr are, respectively, the Reynolds and Prandtl numbers of the fluid defined by

$$Re \equiv \frac{U_c D}{\nu} = \frac{\rho U_c D}{\mu} \quad (5)$$

$$Pr \equiv \frac{c_p \mu}{K} \quad (6)$$

where $\nu \equiv \mu/\rho$ = kinematic viscosity of the fluid,

μ = viscosity of the fluid,

ρ = density of the fluid,

C_p = specific heat of the fluid,

k = thermal conductivity of the fluid,

$U_c \equiv$ corrected velocity of the fluid (corrected for blockage) = $1.22U_a$ for air flow

$U_a \equiv$ free-stream air velocity

(3) Radiation

In contrast to conduction and convection that require a medium for heat transfer, radiation requires no medium for heat transfer. Radiation transfers energy via electromagnetic waves that can travel

through vacuum. The rate of radiant energy transfer from an emitting object with a surface area A_s and a temperature T_s to its surroundings at temperature T_a by is given by:

$$Q_r = \sigma \xi F A_s (T_s^4 - T_a^4) \quad (7)$$

where σ = Stefan Boltzmann constant = $56.7 \times 10^{-9} \text{ W/m}^2 \text{ K}^4$

ξ = the emissivity of the surface (dimensionless)

F = view factor for the emitting object, often assumed to be 1.0.

To conform to the standard form of heat transfer equation, $Q = H A_s \Delta T$, which introduces the heat transfer coefficient H for either conduction and convection, one may analogously define a radiation energy transfer coefficient H_r such that the standard form of the heat transfer equation also applies to radiation energy transfer, i.e.

$$\sigma \xi F A_s (T_s^4 - T_a^4) = Q_r \equiv H_r A_s (T_s - T_a) \quad (8)$$

which requires that

$$H_r \equiv \frac{\sigma \xi F (T_s^4 - T_a^4)}{(T_s - T_a)} = \sigma \xi F (T_s^2 + T_a^2)(T_s + T_a) \quad (9)$$

Unlike the heat transfer coefficients for conduction and convection, which have some theoretical basis, the energy transfer coefficient H_r defined for radiation does not have a sound theoretical basis. It is essentially a fitting parameter with no predictive capability. Equation 9 indicates that H_r will vary greatly with temperature.

Method Designed for Studying Convection and Radiation Heat Transfer

A horizontal metal cylinder of known diameter and length is heated electrically at a known adjustable rate while being exposed to air under varying airflow conditions. The cylinder is insulated at both ends and at contact points with support to minimize conduction losses. The voltage and current applied to the cylinder will be measured, along with the air speed and temperatures of the cylinder and ambient air. The contributions of free convection, forced convection, and radiation to the heat loss from the heated cylinder under the various temperature and airflow conditions will be determined by an analysis of the data in conjunction with the prescribed equations for the heat transfer coefficients of the three heat transfer modes.

Equipment Setup

The apparatus is the HT14 Combined Convection and Radiation Apparatus made by Armfield, which is represented by a mimic diagram on a computer screen as shown in Figure 1. Air enters the vertical duct from the bottom of the apparatus, flowing upward toward a heated cylinder. After the air is heated, it leaves the top of the apparatus and enters the atmosphere. During trials testing forced convection, a fan is used to drive air over the heated cylinder; the fan is not used in trials testing free convection. Air velocity (U_a) is measured using an anemometer located in the middle of the duct, while the temperature of the heater (T_s) and inlet air (T_a) are measured by thermocouples.

The HT14 apparatus is supported by HT10XC Heat Transfer Service Unit, which is an orange color box with dials and switches and which provides an option for manually setting and recording the experimental parameters. Furthermore, there is a remote option through an USB connection that permits the Service Device Unit to be hooked up to a Windows 10 computer so that the apparatus can be monitored and controlled from the computer (via the mimic diagram, see **Section (2)** under **Procedure**).

Procedure

(1) Pre-Experiment Check

Familiarize yourself with the HT14 Combined Convection and Radiation Apparatus, the HT10XC Heat Transfer Service Unit.

Ensure that the horizontal cylinder on the HT14 apparatus is located at the top of the stainless steel duct, with the thermocouple located on one end of the cylinder. If necessary, loosen the thumb screw on the mounting support that holds the cylinder and rotate the cylinder to adjust the dotted angular scale indicator to -90° . Ensure that the thumb screw is securely tightened after any adjustment.

Ensure that the USB cable is connected to both the HT10XC service unit and the computer.

Also ensure the following: (1) the thermocouple that is attached to the heated cylinder is connected to the socket T10 on the front of the HT10XC service unit, (2) the thermocouple that is located in the vertical duct is connected to the socket T9 on the service unit, (3) the main lead from the fan (terminated at the connection box alongside the fan) is connected to the socket marked Output 1 at the rear of the HT10XC, (4) the lead from the anemometer in the vertical duct is connected to the socket marked U_a on the front of the service unit, and (5) the service unit is plugged onto a wall socket.

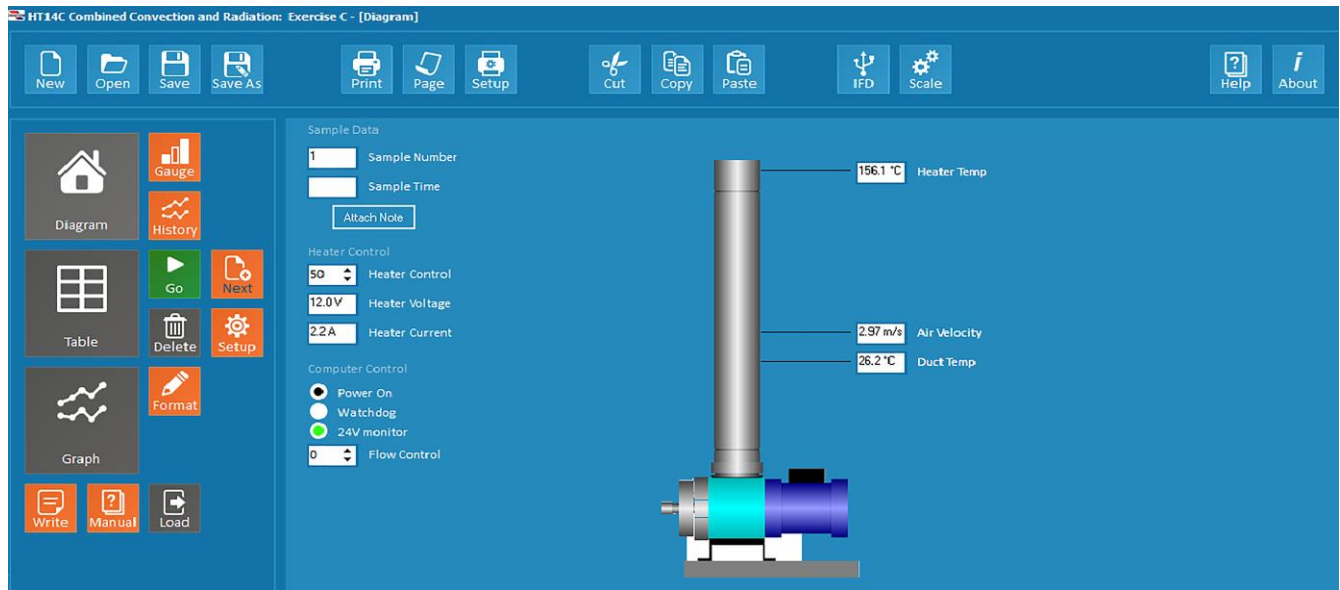


Figure 1. Mimic diagram of experimental apparatus.

Toggle the LCB circuit breaker switch to OFF; the fan will not be used in the study of natural convection but will be used in the study of forced convection. Open the throttle plate at the front of the fan to allow air to enter the fan casing, by turning the adjusting knob anticlockwise.

Set the VOLTAGE CONTROL potentiometer to minimum (anticlockwise) and the selector switch to REMOTE then connect the power lead from the heated cylinder on HT14 to the socket marked O/P 3 at the rear of the service unit.

(3) Experimental Procedure

Turn on the power switch on the HT10 service unit and activate the HT14 software program. Load Exercise C. Click the  button on the toolbar to invoke the mimic diagram of the HT14 apparatus. When the temperature readings displayed for T9 and T10 are stable (they should be identical or very close to each other), click the  button to record the initial readings for T9 and T10, along with that for V, I, and U_a, which should each equal to 0.

Experimental trials should be completed using the optimized run order shown in Table 1; the specified run order minimizes time needed to complete all specified testing by reducing the difference in temperature between data points. To begin, set the heater voltage on the mimic diagram to 20.0 volts by scrolling the up and down arrows on the dialog box for the heater control at 83. Note that the number entry on the heater control box is NOT the actual heater voltage but it is proportional to the heater voltage. The actual heater voltage is shown below the heater control box with the V unit. The readings for heater current, air flow velocity U_a, ambient air temperature

in the duct T9, and the surface temperature of the heated cylinder T10 are displayed on the mimic diagram. When the T10 reading stabilizes, click the  button to record the readings of all five parameters: T9, T10, V, I, and Ua in Table on the toolbar. Then click the  button for the next run.

Repeat the above procedure for each trial specified in Table 1. Avoid setting the heater voltage in excess of 20 V when studying natural convection; otherwise the life of the heating element that is embedded in the metal cylinder will be sharply reduced at an excessively high temperature. For trials investigating forced convection and radiation (e.g. when applied air velocity > 0 m/s), turn on the fan on the HT14 apparatus. Click the  button on the toolbar to invoke the mimic diagram of the HT14 apparatus. Adjust the screw on the throttle plate to achieve the desired airflow velocity (Ua) of 1.0 m/s. A continuous adjustment of the throttle plate screw may be needed to maintain a constant Ua value. When the system reaches a steady state, as indicated by stable readings for the five parameters being monitored (in particular, T10), click the  button to record the readings of T9, T10, V, I, and Ua.

Data Analysis

The dimensions of the heated cylinder are given as $D = 0.010\text{m}$ and $L = 0.070\text{ m}$. Therefore, the heat transfer area for the heated cylinder is $A_s = \pi DL = 0.0022\text{ m}^2$ The emissivity of the cylinder surface is given as $\xi = 0.95$. The view factor F for the heated cylinder is assumed to be 1.00.

- Rate of electrical heat input to the cylinder is given by $Q_{in} = V \cdot I$
- Total rate of heat loss from the cylinder is given by $Q_{Tot} = Q_c + Q_r$
- Rate of heat loss due to natural convection is given by $Q_c = H_c A_s (T_s - T_a)$

Table 1. Optimized trial order for Convection and Radiation experiment.

RUN #	Voltage (V)	Air Velocity (m/s)
1	20	0
2	20	1
3	15	0
4	20	3
5	12	0
6	15	1
7	20	5
8	20	7
9	12	1
10	15	3
11	8	0
12	15	5
13	15	7

14	12	3
15	12	5
16	8	1
17	12	7
18	5	0
19	8	3
20	8	5
21	5	1
22	8	7
23	5	3
24	5	5
25	5	7

- Rate of heat loss due to radiation is given by $Q_r = H_r A_s (T_s - T_a)$
 - Heat transfer coefficients for natural convection, forced convection, and radiation (H_{nc} , H_{fc} , and H_r) can be calculated from the data using Equations 2, 4-6 and 9.
- 1) Plot on a **single** figure H_{nc} and H_r as a function of T10 reading in K. Determine the cross-over temperature at which H_{nc} and H_r are numerically equal. Discuss important trends and conclusions.
 - 2) Plot on a **single** figure H_{fc} and H_r as a function of T10 reading in K for the various tested airflow velocities. For each airflow velocity, determine the cross-over temperature at which H_{fc} and H_r are projected to be numerically equal by interpolating/extrapolating based on the appropriate linear and non-linear fits of data trends. Consider the uncertainties of projected cross-over temperatures. Discuss important trends and conclusions.
 - 3) Compare the calculated heat transfer rate due to convection Q_c (Q_{nc} and Q_{fc}) with the calculated heat transfer rate due to radiation Q_r for varying voltages and air velocities. Discuss important trends and conclusions.
 - 4) Compare Q_{in} and Q_{Tot} amongst a selection of trials to demonstrate any differences. Discuss important trends and conclusions.

References

1. W. H. McAdams, Heat Transmission, 3rd edn., McGraw-Hill, New York (1959).
2. S. W. Churchill and M. Bernstein, "A Correlating Equation for Forced Convection from Gases and Liquid to a Circular Cylinder in Cross Flow", *J. Heat Transfer*, 99, 300-306 (1977).